$$= \frac{1}{8}\left[(2y)_0^2 + (2x)_0^2\right]$$

$$= \frac{1}{8}[(4-0)+(4-0)]$$

$$= \frac{1}{8} \times 8 = 1$$

Therefore, $f(x, y)$ is a probability density function.

Example 9.8

Joint p.d.f. of X, Y is $f(x, y) = \frac{3}{2}\begin{cases} x^2 y & , \ 0 \le x \le 1, 0 \le y \le 2 \\ 0 & , \ elsewhere \end{cases}$ then find the marginal density function of X and Y.

Solution:

$$f(x) = \int_{-\infty}^{\infty} f(x, y)dy$$

$$f(x) = \frac{3}{2}\int_0^2 x^2 y dy$$

$$= \frac{3}{2}x^2\left[\frac{y^2}{2}\right]_0^2 = \frac{3}{2}x^2\left[\frac{4}{2} - \frac{0}{2}\right]$$

$$= \frac{3}{2}x^2 \times 2 = 3x^2$$

$$f(y) = \int_{-\infty}^{\infty} f(x, y)dx$$

$$f(y) = \frac{3}{2}\int_0^1 x^2 y dy$$

$$= \frac{3}{2}y\left(\frac{x^3}{3}\right)_0^1 = \frac{3}{2}y\left(\frac{1}{3}\right) = \frac{y}{2}$$

Marginal density function of X

$$f(x) = \begin{cases} 3x^2 & , \ 0 \le x \le 1, \\ 0 & , \ elsewhere \end{cases}$$

Marginal density function of Y

$$f(y) = \begin{cases} \frac{y}{2} & , \ 0 \le y \le 2, \\ 0 & , \ elsewhere \end{cases}$$

Example 9.9

Joint p.d.f. of X, Y is $f(x, y) = \begin{cases} \frac{1}{10}(4x - 2y) & , \ 1 \le x \le 3, 1 \le y \le 2 \\ 0 & , \ elsewhere \end{cases}$. Find the marginal density function of X and Y